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# SSM Optimization

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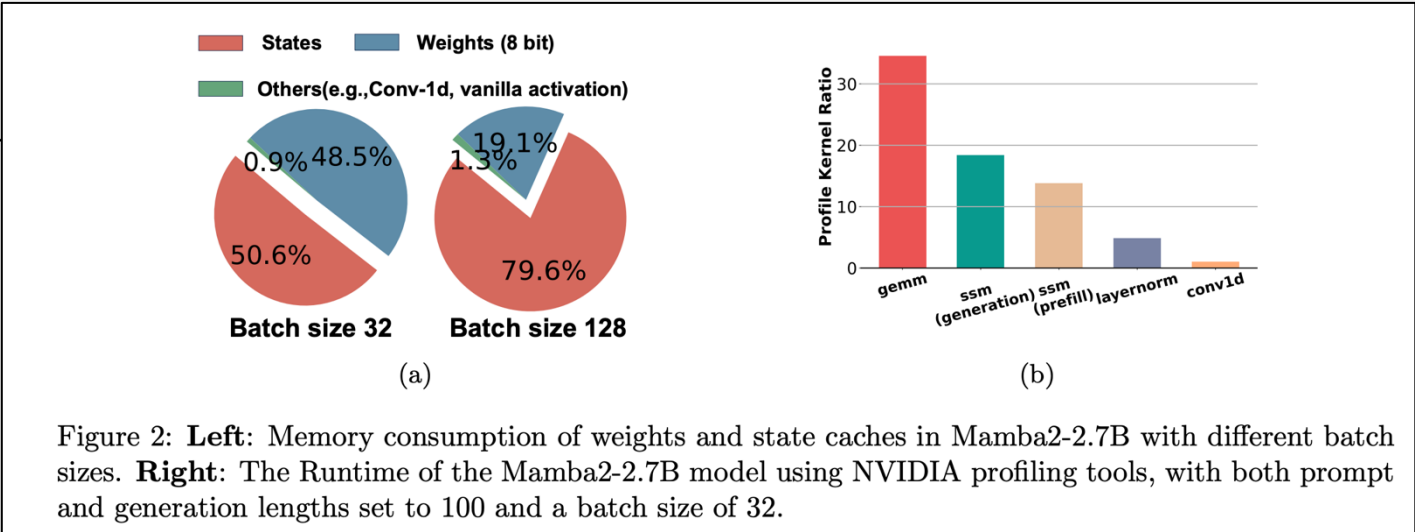
Weekly Meeting

17/07/2026

# State impact

|                      | Falcon Mamba 7B       |                       | Mamba2-2.7B           |                       |
|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                      | BS = 32               | BS = 128              | BS = 32               | BS = 128              |
| Trainable parameters | (96.4%) 7,272,665,088 | (87.1%) 7,272,665,088 | (66.8%) 2,702,599,680 | (33.5%) 2,702,599,680 |
| State values         | (3.6%) 268,435,456    | (12.9%) 1,073,741,824 | (33.2%) 1,342,176,000 | (66.5%) 5,368,704,000 |

Now comparable assuming  
W8 and A/states @ 16 bits



# On State Reduction in Linear Attn.

- Relative readout error is governed by the state rank utilization (ratio between effective rank and theoretical max rank)
  - Low utilization → more sensitivity to noise
- They prune jointly keys and queries
  - They make their strategies compatible with depthwise convolutions
- They do recovery fine-tuning

# On State Reduction in Linear Attn.

- Results for Mamba-2 370M:

Table 4. Mamba-2 370M at 75% state reduction (128  $\rightarrow$  32). Zero-shot accuracy and Wikitext perplexity (pre-RFT).

| METHOD   | ARC-E | ARC-C | HELLAS | WINO | PIQA | WIKI $\downarrow$ |
|----------|-------|-------|--------|------|------|-------------------|
| BASELINE | 49.8  | 27.1  | 40.2   | 52.4 | 65.7 | 29.2              |
| GRAD     | 47.0  | 26.7  | 36.8   | 50.6 | 64.1 | 44.6              |
| DRRQR    | 44.2  | 26.1  | 36.1   | 51.0 | 63.2 | 47.3              |
| S-WANDA  | 41.7  | 25.3  | 35.4   | 53.0 | 62.2 | 52.8              |
| L1       | 40.3  | 24.9  | 34.6   | 51.7 | 62.0 | 54.3              |
| RAND     | 41.0  | 25.0  | 34.7   | 50.6 | 61.5 | 52.8              |

# Next steps

- Profiling tests with simpler operations to debug the roofline model with the profiled results
- From previous point results, consider activations quantization & state compression (and gather stats from more architectures as well)